INCH-POUND

MIL-PRF-32753
14 September 2023

PERFORMANCE SPECIFICATION

Finished Material Properties for Additive Manufactured Fabrication of Low Alloy Steel Components

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of finished material properties using an ultra-high strength, steel powder, feedstock material to establish a successful build by additive manufacturing with pass/fail, performance-based criteria for fabricated, high-quality parts for use on systems and support equipment.

1.2 Classification. The technique will be of the following types, as specified (see 6.2).

1.2.1 Type I. Laser Powder Bed Fusion.

1.2.2 Type II. Powder-Directed Energy Deposition.

Comments, suggestions, or questions on this document should be addressed to Director, DEVCOM U.S. Army Research Laboratory (ARL), Army Research Directorate, DEVCOM ARL Specifications & Standards Office, ATTN: FCDD-RLA-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information may change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9630

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are referenced in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Single copies of this specification, and other federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, https://www.gsa.gov/).

(Copies of this document are available online at https://quicksearch.dla.mil/.)

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-32667 - Ultra High Strength Steel Powder for Additive Manufacturing Fabrication of Parts

(Copies of this document are available online at https://quicksearch.dla.mil)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8 / E8M - Standard Test Methods for Tension Testing of Metallic Materials

ASTM E18 - Standard Test Methods for Rockwell Hardness of Metallic Materials

ASTM E23 - Standard Test Methods for Notched Bar Impact Testing of Metallic Materials

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ASTM F2971 - Standard Practice for Reporting Data for Test Specimens Prepared by Additive Manufacturing

ASTM F3187 - Standard Guide for Directed Energy Deposition of Metals

(Copies of these documents are available from https://www.astm.org).

International Standards

ISO/ASTM 52921 - Standard Terminology for Additive Manufacturing - Coordinate Systems and Test Methodologies

(Copies of this document are available from https://www.astm.org).

(ISO Standard: Available from American National Standards Institute (ANSI), https://www.ansi.org).

Society of Automotive Engineers International Standards

SAE Aerospace Material Specification

AMS-7003 - Laser Powder Bed Fusion Process

(Copies of this document are available from https://www.sae.org).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. All powders used shall be as specified in the contract or purchase order and supplied under MIL-PRF-32667 ultra-high strength steel powder specification for chemical composition and quality program requirements (see 6.2).

3.2 Stress relief and heat treatment technique. Post deposition stress relief and subsequent heat treatment shall be such as to enable the components to meet the requirements of these specifications. Additional requirements shall be negotiated between the procuring activity and the supplier.

3.3 Mechanical testing orientation. Mechanical test specimens shall be machined from bars or built to near net shape and built such that they are tested in the X, Y orientation in accordance with ISO/ASTM 52921 or other orientations as agreed upon procuring activity and supplier.

3.4 Mechanical properties. The finished build of types I and II shall meet the following minimum mechanical property requirements before (as built) and after any post-processing. If mechanical

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property requirements differ from those contained in table I or if any other properties are required, the requirements shall be negotiated between the procuring activity and the supplier.

3.4.1 Minimums. Table I lists the minimum mechanical property requirements by type and test. The types listed in table I shall meet the requirements when tested as outlined for the following tests in 4.8.1, 4.8.2, 4.8.3, 4.8.4, and 4.8.5.

Table I: Minimum mechanical property requirements

Type	Yield Strength (MPa)	Ultimate Tensile Strength (MPa)	Elongation to Failure (%)	Rockwell C Hardness (HRC)	Impact Toughness (ft-lbf)
Type I	1200	1500	18	40	30
Type II	1050	1250	19	40	30

3.5 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.2).

3.6 Workmanship. Component produced under this specification shall be uniform in quality and clean, smooth, and sufficiently free from laminations and other defects which may affect its use. Any additional surface finish requirements shall be specified by procuring activity (see 6.2).

3.7 User instruction markings. Legibly marked or labeled, and any special markings shall be specified by the contract or purchase order (see 6.2).

3.8 Safety Data Sheet (SDS). A SDS for the powder used by this specification shall be included with each shipment and submitted to the end user and contracting activity as stated in FED-STD-313.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements and any supplementary requirements specified herein are classified as follows:

First article inspection and tests (see 4.2).
Quality inspection/Production acceptance (see 4.6).

4.2 First article inspection and tests. When specified (see 6.2), the component supplier shall furnish a test sample or samples for first article inspection and approval. The supplier shall comply with this requirement at the time of the purchase order or contract with the procuring activity and at any time as agreed upon by the part supplier and contracting agency. Test results shall be provided to the procuring activity and shall comply with the purchaser's requirements.

4.3 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2g), the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the

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supplier may use his own ISO 9000/IEC 17025 or NADCAP certified facility or any other facilities certified to ISO 9000/IEC17025 or NADCAP for the performance of the inspection requirements specified herein. Daily inspection events which contribute to the overall monthly OQE need not to be performed by an ISO 9000/IEC 17025 or NAPCAP certified facility. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.4 Inspection conditions. Unless otherwise specified in the purchase order or contract, all test specimens shall be conditioned and tested at standard conditions.

4.4.1 Standard conditions. Unless otherwise stated in the test method or paragraph, standard condition shall be 73 ± 4 °F (23 ± 2 °C) and a 40 – 70 percent relative humidity.

4.5 Quality inspection/Production acceptance. All components shall meet all requirements of section 3 for procedures listed in section 4.8. The inspections set forth in this specification shall become a part of the supplier's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the supplier of the responsibility of ensuring that all components or parts submitted to the Government for acceptance comply with all requirements of the contract or purchase order. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective component, either indicated or actual, nor does it commit the Government to accept defective component.

4.6 Testing responsibility and facilities. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all the requirement tests as specified herein.

TABLE II. Index.

Item	ASTM Method	Test Paragraph	Requirement Table
Yield strength	E8/E8M	4.8.1	Table I
Ultimate tensile strength	E8/E8M	4.8.2	Table I
Elongation to failure	E8/E8M	4.8.3	Table I
Rockwell hardness C	E18	4.8.4	Table I
Impact toughness	E23	4.8.5	Table I

4.8 Test methods - component testing.

4.8.1 Yield strength. The yield strength shall be measured following the procedure specified in ASTM E8 / E8M method (see 3.4.1 and table I). Size, location, gauge length, and orientation as specified for test specimen.

4.8.2 Ultimate tensile strength. The test method shall be as specified in ASTM E8 / E8M. (see 3.4.1 and table I).

4.8.3 Elongation to failure. Test in accordance with ASTM E8 / E8M. (see 3.4.1 and table I).

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4.8.4 Rockwell hardness C (HRC) test. The hardness shall be measured with a meter following the procedure specified in ASTM E18 as agreed upon by the component supplier and procuring activity. The readings shall be taken on the surface and interior. The average value shall be determined for each surface (see 3.4.1 and table I).

4.8.5 Impact toughness. Test in accordance with ASTM E23. (see 3.4.1 and table I).

4.8.6 Heat treatment technique. Processing/post-processing test specimens shall be prepared as specified in ASTM F2971.

4.9 Rejection and retest. Failure in any inspection will result in the rejection of the components from large-scale production. Rejected parts cannot be resubmitted for acceptance without written approval from the contracting activity. The application for resubmission will contain all details concerning previous rejections and measures taken to correct these deficiencies.

4.10 Safety Data Sheet (SDS). The SDS shall address all components of the powder and be prepared in compliance with FED-STD-313. The SDS shall be included with each shipment of the material and forwarded to pertinent Government agencies as stated in FED-STD-313. Noncompliance to 3.8 shall be cause for rejection.

4.11 Manufacturer's certification. The manufacturer shall certify and maintain substantiating evidence (Objective Quality Evidence) that the build meets the requirements of this specification, and that the component conforms to the manufacturer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery or provided for under the provisions of the contract.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that is helpful but is not mandatory.)

6.1 Intended use. The finished material properties covered by this specification is intended for use in the manufacture of components using ultra-high strength steel powder.

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6.2. Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type required (see 1.2)
c. If issues of documents are different (see 2.2.1 and 2.3)
d. Specify type of powdered alloy required (see 3.1)
e. If first article inspection is different (see 3.5 and 4.2)
f. If specific surface finish/roughness requirements are required (see 3.6)
g. Selection of applicable level of markings and packaging requirements (see 3.7 and 5.1)
h. When certification is acceptable (see 4.3 and 6.9)
i. Packaging requirements if other than specified (see 5.1)

6.3 Objective Quality Evidence. A documented statement of fact, or supporting record, either quantitative or qualitative, pertaining to the quality of an item or process, based on observations, measurements, or tests that can be verified as required within the document. Evidence will be expressed in terms of specific quality requirements or characteristics in a certified lab report. These characteristics are identified in drawings, specifications, and other documents, which describe the item, process, or procedure.

6.4 ASTM F3187. Standard Guide for Directed Energy Deposition of Metals for definitions.

6.5 SAE AMS-7003. Laser Powder Bed Fusion Process.

6.6 Performance specification. MIL-PRF-32667 feedstock material publication (document date: 28 October 2020), Ultra High Strength Steel Powder for Additive Manufacturing Fabrication of Parts, is available for sourcing the steel alloy powder and quality control inspection to establish a successful build.

6.7 Cognizant Engineering Authority. The CEA assumes all responsibility for assuring that all required testing and subsequent evaluation of any powders are completed and pass all requirements specified herein, and also meet any additional performance criteria specific to their intended application.

6.8 Engineering Support Activity. The ESA provides technical data, quantities, packaging requirements, possible suppliers, special handling, storage requirements, and shelf-life information to procure the powder and resolve individual engineering support problems.

6.9 Process Approval. When specified in the contract (see 4.3 and 6.2), for types I and II, in lieu of submitting written procedures identifying processes, suppliers certified as specified herein are acceptable. Be prepared to offer evidence of the certification if requested.

a. National Aerospace and Defense Contractors Accreditation program (NADCAP), Performance Review Institute (PRI) industry-managed accreditation programs. See https://www.eauditnet.com.

b. NACE Business Services, LLC, NACE International Institute for Contractor Accreditation Program (NIICAP) accredited contractors. See https://www.niicap.net/.

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c. ISO 9001/2015 or ISO/IEC 17025 Quality Management, Conformity Assessment Accreditation from an external certification body. See: https://www.iso.org/home.html.

In reviewing and validating certification by the Engineering Support Activity (ESA), suppliers certified should submit their accreditation from the external certification body or registrar.

6.10 Additive Fabrication. Method, material, and manufacturing processes.

6.10.1 Method of fabrication. New design documents, engineering drawings, and ordering data should indicate the type of technique required, dataset system requirements, orientation and location, and parameters when practicable.

6.12 Subject term (key word) listing.

Assemblies
Elongation
Hardness test
Heat treatment
Laser
Metal AM
Minimum acceptance
Part testing
Quality
Surface finish
Tensile testing

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CONCLUDING MATERIAL

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Navy - AS

Review activities:
Army – AV, CR
Navy – OS, SH
DLA – CQ, DH

Civil agency:
GSA - FAS

Preparing Activity:
Army - MR

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